After warmup

Linear Decay

The learning rate decrease in a straight line from the peak value down to some minimum (often 0) over the remaining training step.

Formula

After warmup, from step t to total step T

```
lr(t) = base_lr * (1 - (t - warmup_steps) / (T - warmup_steps))
```

So at the end of the warmup, the lr = base_lr and the final step, the lr = 0

Common usage: NLP, BERT-style training

Cosine Decay (Cosine Annealing)

The learning rate follows half a cosine curve, it decrease slowly at first, speeds up in the middle, then slows down again near the end.

Formula

```
lr(t) = min_lr + 0.5 * (base_lr - min_lr) * (1 + cos(π * (t - warmup_steps) / (T - warmup_steps)))
```

So at the end of warmup, the lr = base_lr and at the final step lr = min_lr

Common usage: Large-scale/LLM pretraining (GPT, LLaMA,...)

Why the Cosine decay is preferred in LLM

- Keep the learning rate high for longer early on (more aggressive learning while it's safe to do so)
- Taper very gently at the end (helps the model "settle" carefully into a minimum instead of dropping off a cliff)

  This gentle tapering at the end is often cited as helping final model performance, which is part of why it's the default in many modern LLM training recipes (GPT-family, LLaMA, etc.), usually paired with linear warmup.